# Ball physics in FreeKiki

Drag, the Magnus effect, and the trajectory of a hand-corrected ball, with spin inferred from pixels. *vailá*, version 0.4.11, 26 October 2026.

## 1. Summary

3D reconstruction of a hand-corrected ball needs a physically plausible trajectory. A soccer ball is not a particle on the ground: it leaves the foot, rises, loses speed to the air, and curves because of spin. This document explains the model FreeKiki uses, where the numbers come from, how it is fitted to the markings, and what still does not match.

In short, the model:

- computes drag with a Cd that depends on Reynolds number (the drag crisis);
- computes the Magnus effect that curves the ball, with a lift coefficient that depends on spin;
- fits initial velocity, spin, and kick time to the ball pixels, using each frame's DLT3D calibration;
- uses three physical points (kick foot, barrier passage, and arrival at the goalkeeper's hands) as anchors.

> Honest result: on the c1 clip the model reproduces the ball markings with 1.6 px mean error, but the three physical anchors are only met with 0.6 to 1.0 m error. Validation against the published Pelé kick case also failed (about 10 to 12 m error). Details in section 8.

## 2. Forces on the ball

A ball moves through air under three forces: weight, drag (opposite to the motion), and the Magnus force (perpendicular to the motion, caused by rotation).

### 2.1 Drag

Drag is the force the air exerts against the motion. For a sphere (the ball) it is:

$$F_D = \tfrac{1}{2} \cdot \rho \cdot A \cdot Cd \cdot v^2$$

where $\rho$ is air density (1.2 kg/m³), $A$ is the cross-sectional area ($\pi r^2$, with r = 0.11 m), and $Cd$ is the drag coefficient, which is not constant: it depends on speed and ball roughness.

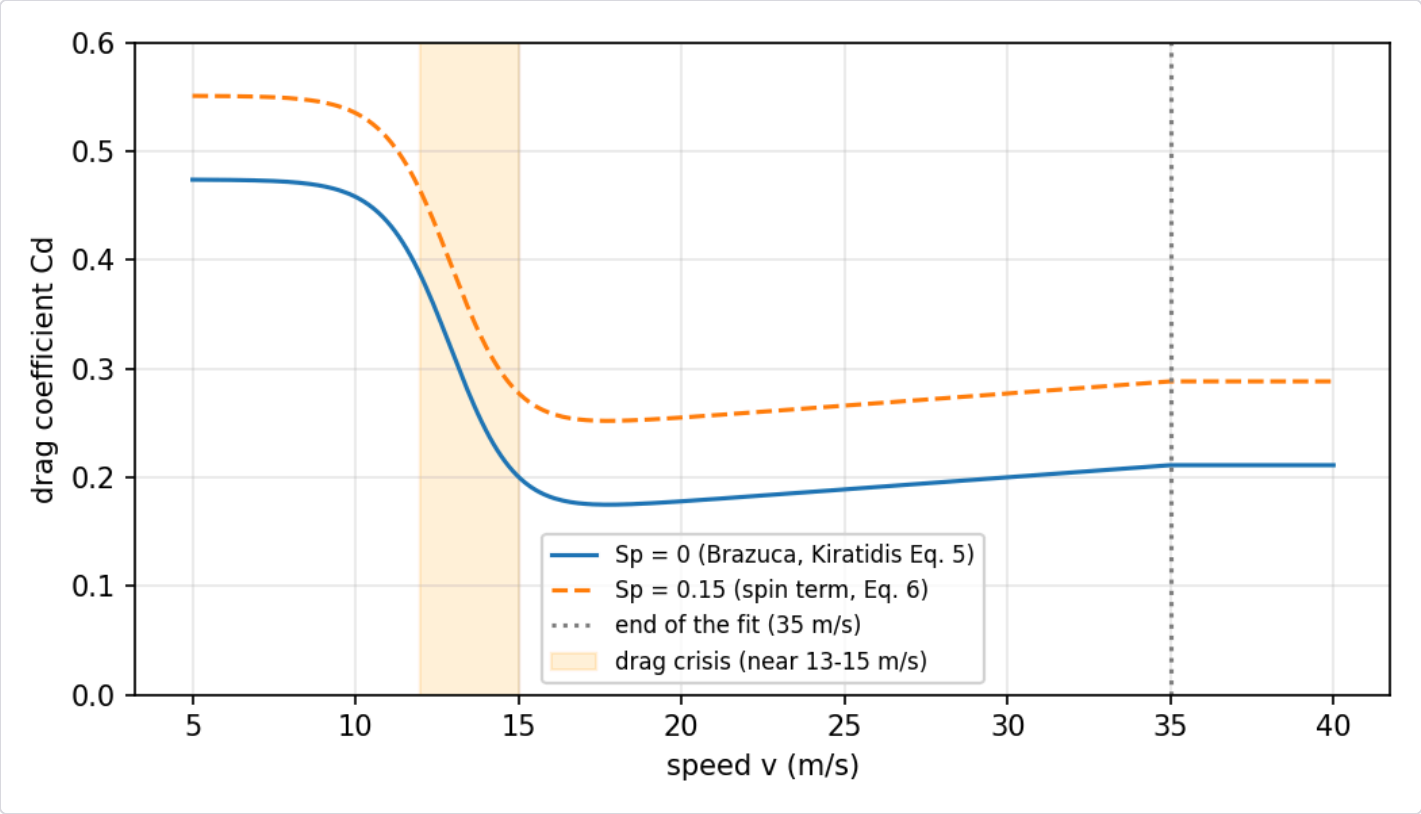
The **Reynolds number** measures the ratio of inertia to air viscosity:

$$Re = v \cdot D / \nu \qquad \nu = \mu / \rho = 1.8 \cdot 10^{-5} / 1.2 = 1.5 \cdot 10^{-5}\ m^2/s \qquad D = 0.22\ m$$

For a soccer ball, Re lies between $10^5$ and $5 \cdot 10^5$ in play: the range where drag changes regime.

### 2.2 The drag crisis

Below a critical Reynolds number (about 2 to $3 \cdot 10^5$ for a soccer ball, after Asai & Seo, 2013), the air boundary layer is laminar and separates early from the ball, leaving a wide wake: Cd ≈ 0.47. Above the critical value the boundary layer becomes turbulent, separates later, and the wake shrinks: Cd falls to 0.12–0.15. That drop is the **drag crisis**. Seams and panels bring the transition forward, so Brazuca and other panelled balls have crises at different speeds.




Brazuca drag coefficient (fit to Table 1 of Kiratidis & Leinweber, 2018), from `freekiki_physics.drag_coefficient`. The orange band marks the crisis.

### 2.3 Magnus effect

A spinning ball drags air on one side and slows it on the other. The pressure difference yields a force perpendicular to velocity and to the spin axis:

```
F_M = ½ · ρ · A · C_L · v² · (ω^ × v̂)
```

The coefficient $C_L$ grows with the **spin parameter** $Sp = r \cdot |\omega| / v$ (ω in rad/s). We use the Kiratidis & Leinweber (2018, Eq. 8) fit: $C_L = 1.15 \cdot Sp^{0.83}$, without the drag factor of Eq. 9 (documented simplification). For a kick at 6 revolutions per second and 30 m/s, Sp ≈ 0.14 and C_L ≈ 0.22.

Spin also increases drag: $Cd \mathrel{+}= b \cdot Sp$ with b = 0.514 (Kiratidis Eq. 6, measured on Teamgeist; Brazuca has no own value in that paper).

## 3. Equation of motion

With z upward, the ball acceleration is:

```
a = −g·ẑ − (ρA / 2m) · Cd(v, Sp) · |v| · v + (ρA / 2m) · C_L(Sp) · |v|² · (ω̂ × v̂)
```

with g = 9.81 m/s², m = 0.43 kg (Kiratidis) and D = 0.22 m. Integration is numerical (RK45, `scipy.integrate.solve_ivp`). Before the kick the ball rests on the ground. From kick time $t_k$ it flies under the equation above.

### 3.1 Constants and sources

| quantity | value | source |
|---|---|---|
| air density ρ | 1.2 kg/m³ | Aguiar (UFRJ); Kiratidis |
| viscosity μ | $1.8 \cdot 10^{-5}$ Pa·s | Aguiar (UFRJ) |
| diameter D, radius r | 0.22 m, 0.11 m | Aguiar; Kiratidis |
| mass m | 0.43 kg | Kiratidis (Asai & Seo give 0.428–0.438 kg) |
| Cd(v), Brazuca | a = 0.4740; $v_c$ = 12.92; $v_s$ = 1.000; $b_{min}$ = 0.1657; $b_{max}$ = 0.2112; $v_{min}$ = 14.61; $v_{max}$ = 35.00 m/s | Kiratidis, Table 1 and Eq. 5 |
| spin term in Cd | b = 0.514 | Kiratidis, Eq. 6 (Teamgeist value) |
| C_L(Sp) | $1.15 \cdot Sp^{0.83}$ | Kiratidis, Eq. 8 (without drag factor) |

> The Cd table goes up to 35 m/s. Above that, the code holds the 35 m/s value and flags the segment in the report (`above_fit_range`).

## 4. How the code fits the trajectory

`getpixelvideo` markings give, for each frame, the ball position in the image (u, v). Each frame has a DLT3D calibration, a 3×4 matrix P that maps a field point (X, Y, Z) in metres to the image. The fit seeks the physical trajectory that, projected by P each frame, reproduces the markings.

Fitted parameters (9):

- initial ground position ($x_0$, $y_0$);
- initial velocity ($v_x$, $v_y$, $v_z$);
- kick time $t_k$;

- spin vector $\omega = (\omega_x, \omega_y, \omega_z)$. Direction and magnitude are inferred: curvature in X,Y points to the vertical spin component, curvature in Z to the horizontal components.

Residuals minimized (`scipy.optimize.least_squares`):

- **pixels**: per-frame reprojection error, 2 px sigma;
- **kick foot** (frame 0): the ball starts at the player's foot (left ankle), 0.3 m sigma;
- **barrier** (frame 9): the ball is 1.8 m high and 0.3 m horizontally from the barrier player's shoulder; 0.1 m sigmas;
- **goalkeeper** (last frame): the ball reaches the midpoint of the keeper's hands, 0.2 m sigma.

The fit is repeated with three kick times and three spin starts (zero, and ±6 rps about the vertical axis) to avoid a local minimum. The lowest-cost run wins.

Usage in FreeKiki:

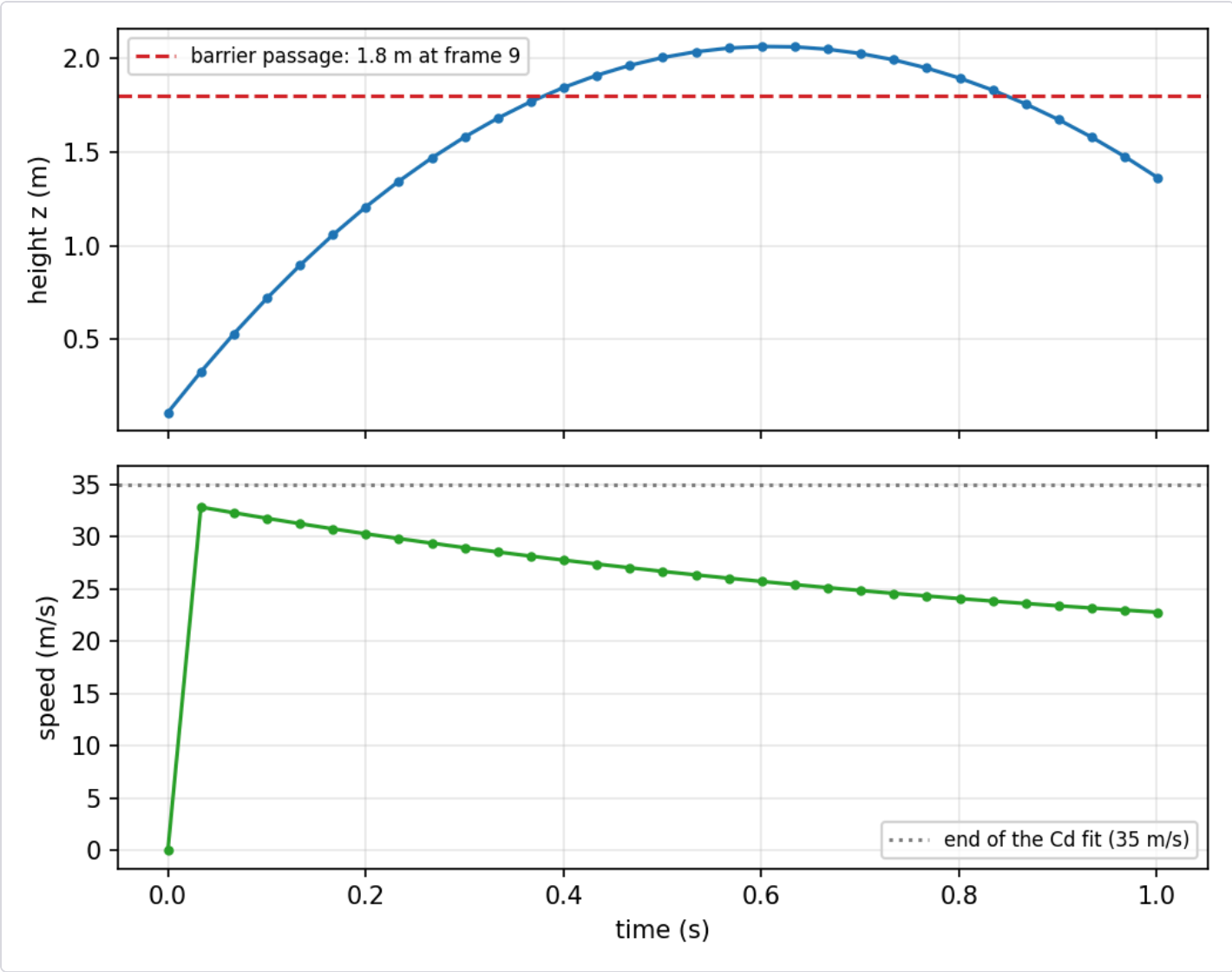
```
uv run --no-sync vaila/freekiki.py ball-rebuild -w WS --out <3D folder> --ball-csv <
markers.csv> \
  --ball-anchors '{"kicker_id": 30, "barrier_id": 15, "keeper_id": 21}'
```

IDs are the players in the clip (`person_id` in the pose files). Output includes `ball_flight.json` with parameters, speeds, and residuals; the ball C3D then carries true height.

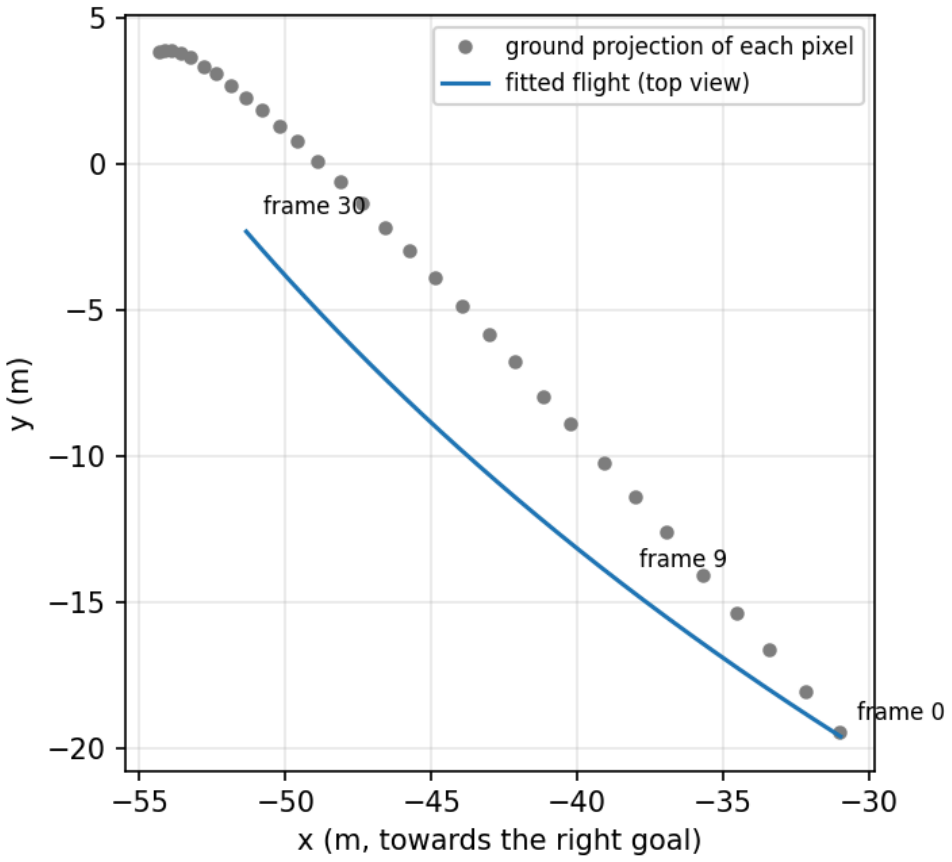
## 5. Results on clip c1 (Paquetá)

Data: 31 frames at 29.97 fps, valid DLT3D on every frame; hand-corrected ball markings.

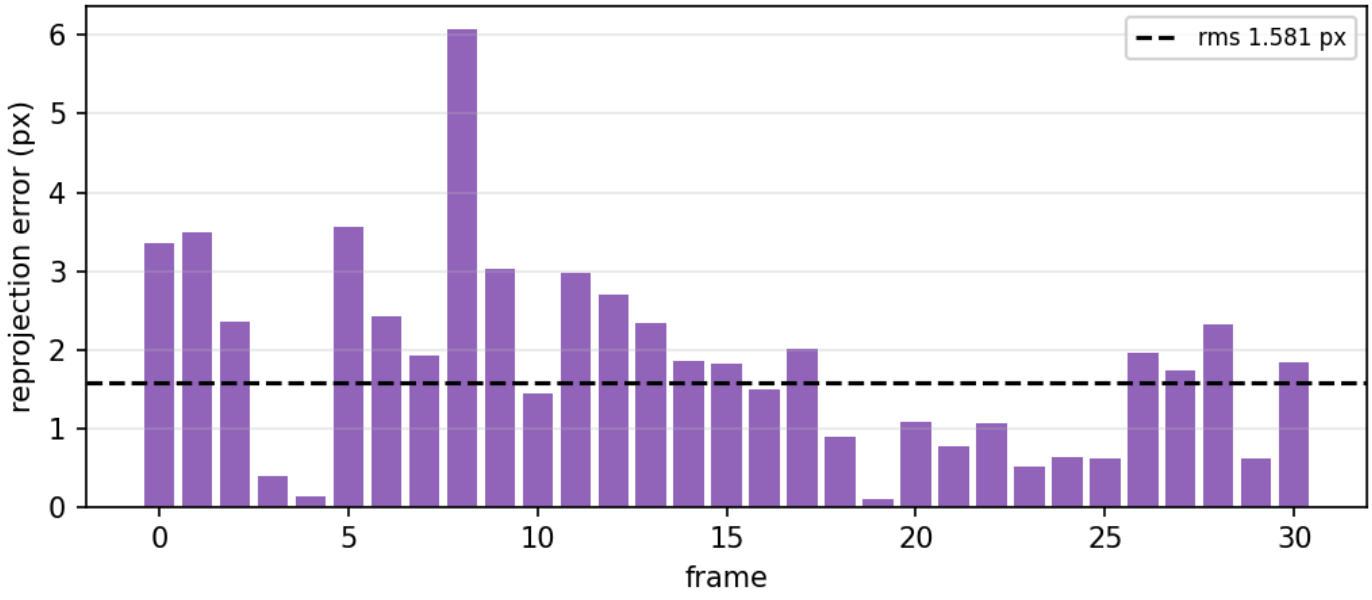
| quantity | result |
|---|---|
| reprojection error (rms / max) | 1.58 px / 6.08 px (the earlier ballistic fit, no Magnus, with anchors, had 8.7 px) |
| kick time | 0.00 s (lower bound of the interval: the ball leaves the foot at frame 0) |
| inferred spin | 4.9 rps, axis (0.08; −0.04; −1.00): nearly vertical, hence the lateral curve |
| peak speed | 32.8 m/s (inside the Brazuca Cd range) |
| speed at end (frame 30) | 22.8 m/s: 30% drop from drag |
| height at frame 9 | 1.58 m (the barrier asks for 1.8 m) |
| maximum height | 2.07 m |
| anchor errors | foot: 0.63 m; barrier: 0.70 m horizontal and 0.22 m height; keeper: 1.02 m |




Height (top) and speed (bottom) of the fitted ball. Speed falls continuously under drag.




Top view. Grey points are the ground projection of each pixel, which assumes the ball on the ground and therefore errs in flight. The blue curve is the fitted track, which matches the pixels within 1.6 px.




Reprojection error per frame, in px. Error stays low through the whole flight.

## 6. Validation

### 6.1 The integrator

With no drag and no Magnus, the simulation matches the analytic oblique-projectile solution ($x = v_x t$, $z = v_z t - g\,t^2/2$) to a maximum error of $6 \cdot 10^{-14}$ m. The integrator is correct.

### 6.2 Recovery of a known trajectory

A trajectory with 5 rps spin was projected through a known camera with 0.5 px noise. The fit recovered every frame's position within 0.25 m and the spin within 10%. That test lives in the repository (`tests/test_freekiki_physics.py`).

### 6.3 Pelé's kick (Aguiar, UFRJ)

Aguiar's notes publish the start and end of a kick: from (−5.2; −2.9; 0) m with v = (27.8; −0.4; 8.8) m/s to (54.3; 3.7; 0) m at t = 3.20 s, with spin −6.11 Hz. Simulating with our model:

| setup | error to the published point |
|---|---|
| Brazuca, most favourable spin axis | 12.0 m |
| Aguiar's own coefficients (Cd = 0.15; $C_S = 1.25 \cdot Sp$) | 9.7 m |
| Brazuca with $C_S = 1.25 \cdot Sp$ | 9.9 m |

> This validation **did not pass** (target was 2 m). The integrator is correct (6.1), so the gap comes from elsewhere: axis and spin-sign conventions in Aguiar's text, how the end point is reported, or a fit specific to the published kick. With the available data those causes could not be separated.

## 7. What was simplified

- **Lift**: $C_L = 1.15 \cdot Sp^{0.83}$, without the drag factor (Kiratidis Eq. 9).
- **Spin in drag**: b = 0.514 is the Teamgeist measurement; Brazuca has no own value in the paper.
- **Speed range**: the Cd table goes to 35 m/s. Above that we keep the 35 m/s value.
- **Bounce and contact**: flight is a single segment. Ground bounces and barrier contact are not modelled.
- **Single camera**: depth is weak. Physical anchors fix that depth, so their quality weighs on the result.

## 8. Limitations of the c1 result

Pixels are reproduced within 1.6 px, and speed and spin are of a plausible order. But the three anchors are not satisfied together: the foot is 0.63 m off, the barrier 0.70 m (horizontal), and the keeper 1.02 m. Because the fit weights the 31 pixel frames more than the three anchors, it chooses the trajectory that

matches the markings. That points to at least one of three problems: this clip's DLT3D (scale or camera pose, about 1 m), the frames chosen for the anchors (frame 9 and the last frame), or the players' field positions. To decide, the clip still needs a visual check with the trajectory overlaid.

Kick time also sits on the bound (0.00 s). At one-frame resolution that is consistent with a kick at frame 0.

## 9. References


1. Kiratidis, A. L.; Leinweber, D. B. *An Aerodynamic Analysis of Recent FIFA World Cup Balls*. arXiv:1710.02784v2, 2018.
2. Asai, T.; Seo, K. *Aerodynamic drag of modern soccer balls*. SpringerPlus 2:171, 2013.
3. Asai, T.; Seo, K.; Kobayashi, O.; Sakashita, R. *Fundamental aerodynamics of the soccer ball*. Sports Engineering 10, 101, 2007 (cited by Aguiar).
4. Aguiar, C. E. *Aerodinâmica da bola de futebol: da Copa de 70 à Jabulani*. UFRJ (lecture notes, futebol2.pdf).
5. NASA, Glenn Research Center. *Drag* (student aerodynamics page, drag equation and Reynolds).
6. Goff, J. E.; Carré, M. J. *Trajectory analysis of a soccer ball*. American Journal of Physics 77, 1020, 2009.


> Figures come from `docs/fisica_bola_figuras.py`, which reads the reconstructed folder and does not re-fit.